

How Perceived Lack of Benevolence Harms Trust of Artificial Intelligence Management

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As organizations continue to supplement and replace human management with artificial intelligence (AI), it is essential that we understand the factors that influence employees' trust in AI management. Across one preregistered field study, where we survey 400 delivery riders in Mainland China, and three preregistered experiments (total $N = 2,350$), we find that AI management is perceived as less benevolent than human management. Given that benevolence is an important antecedent of trust in leaders, this perception has a negative effect on trust in AI management, even when controlling for perceived ability and integrity. Employees prefer human management to AI management in high empathy demand contexts, where individuals seek management that can empathize and experience the emotions that they are feeling, as opposed to low empathy demand contexts. These findings deepen our understanding of trust and provide important theoretical and practical insights on the implementation and adoption of AI management.

Keywords: trust, benevolence, emotions, artificial intelligence management

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Companies are increasingly integrating artificial intelligence (AI)-powered systems to oversee worker productivity, manage extensive operations, and enhance organizational efficiency (Kaplan et al., 2023; Logg, 2019), with the International Data Corporation (IDC) predicting that 80% of Global 2,000 companies will adopt AI managers to hire, fire, or train employees within a few years (Mearian, 2022). While proponents of such systems highlight the potential of AI management to improve organizational competitiveness and workforce well-being (e.g., reduce injuries), employees managed by AI, such as those at Amazon's warehouses, have expressed feeling like “disposable cogs” in an “unforgiving” workplace (Gurley, 2021; Rad, 2022; Soper & Day, 2021). Despite AI technology's profound capabilities, there are important concerns from employees surrounding its deployment to managerial roles that could limit its adoption and acceptance.

Research on the adoption of technology within organizations has historically focused on its perceived usefulness and ease of use (technology acceptance model; F. D. Davis, 1989). As AI has begun to occupy organizational positions traditionally filled by humans,

recent research has shifted toward a sociotechnical perspective that uses paradigms traditionally reserved for human-to-human interactions to understand human-to-AI interactions (de Visser et al., 2016; Kong et al., 2023; Lazar & Nelson, 2023). For example, Langer et al. (2023) argued that as AI assistants are increasingly used to support decisions with ethical implications such as personnel recruitment, decisions on whether to adopt technology have moved beyond traditional performance measures to questions regarding the morality and fairness of the system's decisions. Trust has been found to provide a powerful paradigm for capturing perceptions toward AI and assessing whether individuals will adopt it (Glikson & Woolley, 2020), and scholars such as Choung et al. (2023) have found that positive attitudes toward AI assistants (e.g., viewing the technology as “pleasant”) are positively correlated with trust of the assistant.

Understanding if employees will trust and accept AI management systems requires examining their perspective on being subservient to AI decisions (Cratsley & Fast, 2024), but extant research provides equivocal findings regarding trust in algorithmic management.

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Dietvorst et al. (2018) found that users displayed strong algorithm aversion when they were unable to make changes to algorithmic decisions, even when the algorithm outperformed humans, which suggests that individuals dislike lacking autonomy and being subservient to algorithmic systems. Similarly, Tong et al. (2021) found that AI feedback systems improve employee performance relative to human feedback systems, but the effect is reversed upon the disclosure of the AI feedback system's presence. In contrast, Raveendhran and Fast (2021) found that privacy concerns lead employees to prefer being closely monitored by an AI system over a human. The results of prior research underscore that understanding trust of AI management systems requires examining employees as the subordinates of AI, indicate that previously unexamined mechanisms might enrich our understanding of adoption of AI management, and suggest that contextual factors play an important role in this relationship.

Within human-to-human interactions, perceived benevolence and emotional context are foundational antecedents of trust (R. B. Lount & Pettit, 2012; Pirson & Malhotra, 2011; White et al., 2024). A vast literature has found that whether an individual trusts a counterpart is largely predicated on beliefs about whether the counterpart cares about the individual and wants to help them (Mayer & Davis, 1999; Mayer et al., 1995), and that this relationship can be strengthened or attenuated based on the emotional context and factors unrelated to the counterpart (e.g., incidental emotions such as anger; Dunn & Schweitzer, 2005; van Kleef & Lelieveld, 2022). While scholars have started to examine the perceived benevolence of AI, they have either compared the perceived benevolence of AI agents to other AI agents, rather than to humans (e.g., Bedué & Fritzsche, 2022; Choung et al., 2023; W. Wang & Benbasat, 2007), or they have examined the perspectives of humans who are voluntary users, instead of subordinates, of AI decision making (e.g., Bedué & Fritzsche, 2022; Choung et al., 2023; Langer et al., 2023; W. Wang & Benbasat, 2007). Yet, prior work suggests that people are uncomfortable with machines making ethical decisions (e.g., who lives or who dies in driving, medical, and military contexts) because algorithmic agents lack human minds and emotional experience (Bigman & Gray, 2018). If this lack of emotional experience leads employees to believe AI managers do not care about doing good to humans and lack benevolence, then employees may be more reticent to accept AI replacing human managers within organizations, especially in contexts where empathy is essential, which could have negative downstream consequences such as lower employee productivity, decreased organizational commitment (e.g., decreased organizational citizenship behaviors), and higher rates of employee turnover (Brower et al., 2009; Dirks & Ferrin, 2002; Lehmann-Willenbrock et al., 2013; Li & Tan, 2013; Purba et al., 2016).

In this research, we consider if AI management is viewed as less benevolent than human management, the detrimental effect this might have on trust, and if this potential negative relationship is exacerbated in contexts that demand empathy, where employees will be more sensitive to AI's incapability to experience emotions. In doing so, we make several contributions to the literature on trust, emotions, and algorithms and AI adoption. First, we extend interpersonal trust theory (Mayer & Davis, 1999; Mayer et al., 1995) to explain the emerging leader-subordinate relationship between machine and human by considering the perspective of employees who are the subjects of AI management. Second, we draw insights from the emotions as social information (EASI) model (Van Kleef,

2008; van Kleef & Côté, 2022; see also Elfenbein, 2023; Scherer & Moors, 2019) to shed light on the important role that perceived benevolence plays in employee trust of AI management. Third, we contribute to the growing research on AI adoption in the workplace by highlighting the important role that emotional context plays in the acceptance of AI management.

Theory and Hypotheses

Trust

Trust is essential for building effective working relationships, especially supervisor-subordinate relationships (Lewicki & Bunker, 1996; Lewicki et al., 2006; Mayer et al., 1995; S. Wang et al., 2010; Weiss et al., 2021). Trust in supervisors enables subordinates to achieve positive behavioral (e.g., performance and organizational citizenship behaviors) and attitudinal outcomes (e.g., job satisfaction and organizational commitment; Dirks & Ferrin, 2002; Konovsky & Pugh, 1994; Rich, 1997). In contrast, a lack of trust harms organizations by eroding relationships at the dyadic and group level. In line with prior research, we define trust as the willingness to be vulnerable to a counterpart based on positive expectations of the counterpart (Mayer et al., 1995; Rousseau et al., 1998).

Supervisors are more likely to gain the trust of their subordinates if they are perceived to have a high level of ability, benevolence, and integrity (ABI). Ability refers to a trustee's competencies and skills to exert influence in a specific domain (Cook & Wall, 1980; Tomlinson et al., 2020). For supervisors, this encompasses both their formal and informal influence in the organization, along with their recognized competencies and talents (Jones et al., 1975).

Benevolence refers to a trustee's positive orientation toward the trustor and desire to do good to the trustor (Mayer & Davis, 1999; Mayer et al., 1995). The perception of benevolence often depends on the extent to which the trustee offers help to the trustor out of altruism rather than self-interest (Larzelere & Huston, 1980; Strickland, 1958). When supervisors are perceived to sincerely care about their employees' interests and welfare, regardless of extrinsic rewards, they are seen as more benevolent (Dirks & Ferrin, 2002; R. B. Lount & Pettit, 2012).

Integrity refers to a trustee's consistent adherence to a set of trustor-approved principles (McFall, 1987). Supervisors are often considered to have a high level of integrity if they have a strong sense of justice, their values are accepted and endorsed by their employees, and their actions are consistent with their words (Butler, 1991; Butler & Cantrell, 1984). Importantly, integrity is not only determined by the trustee's devotion to the same set of principles as the trustor, but also by the trustee's consistent actions in accordance with the moral codes they claim to uphold.

The prevailing conceptualization of trust highlights the willingness of the trustor to be vulnerable to exploitation (Levine et al., 2018; Lewicki & Bunker, 1996; Lewicki et al., 2006; Rousseau et al., 1998). Building on this conceptualization, the majority of the existing research exploring trust of management has focused on human-to-human interactions, the willingness to be vulnerable to another individual, and the degree to which subordinates view human managers as competent, benevolent, and possessing integrity (Mayer et al., 1995; W. Wang et al., 2016).

Modern organizations are shifting more toward AI management and implementing AI in roles traditionally occupied by human managers, and a burgeoning literature highlights the importance of understanding human-to-algorithm interactions for understanding trust within organizations (Choung et al., 2023; Glikson & Woolley, 2020; Langer & Landers, 2021). For example, W. Wang and Benbasat (2007) found that transparency (explanations of how and why a system made a recommendation) could elevate the extent to which an automated recommendation agent was perceived as competent and benevolent, and increase participants' willingness to use the recommendations. More recently, Bedué and Fritzsche (2022) conducted qualitative interviews with 12 experts in the field of AI (including engineers, managers, information systems specialists, and researchers) who thought perceptions of AI's ability, benevolence, and integrity would be helpful for understanding its risks and benefits.

In this work, we extend interpersonal trust theory and adopt a sociotechnical perspective to examine employees' perceptions of AI management in organizations. Rather than examining humans as the users of technology (e.g., Bedué & Fritzsche, 2022; Choung et al., 2023; Langer et al., 2023; W. Wang & Benbasat, 2007), we consider the perspective of employees managed by AI to deepen our understanding of the social and emotional aspects of human-to-algorithm interactions. Our research advances theoretical and practical understanding of AI management and trust by considering the unique role that benevolence plays in this relationship.

AI Management and Benevolence

We define AI management as a set of computational procedures that serve a management role and make executive decisions that influence target individuals (e.g., work assignment and performance appraisal; Broussard, 2018; Felten et al., 2021; Lee et al., 2015; Logg et al., 2019; Rogers, 1987; Yu et al., 2017). AI management is increasingly common within organizations, taking on responsibilities such as recruitment and performance evaluation (Biswas, 2021; Brynjolfsson & McAfee, 2017; Cheng & Hackett, 2021; Marr, 2018; Raisch & Krakowski, 2021; Simonite, 2015). We anticipate that the perceived benevolence of AI management, versus human management, will influence trust and acceptance of these systems, and we consider the unique role that emotions will play in this relationship.

Emotions have a profound effect in coordinating social interactions and behaviors in the workplace (Elfenbein, 2023; Gaertig et al., 2019; Seger et al., 2017; Van Kleef et al., 2009, 2012). The emotions as social information (EASI) model points to the functionality of emotions in the workplace (Van Kleef, 2008; van Kleef & Côté, 2022); employees can gather information and feedback on how they are perceived and evaluated by others, and thus appropriately adjust their behaviors to solve the problem at hand and improve their performance and success (Fridlund, 2014; Homan et al., 2016). For example, van Kleef et al. (2019) found that coaches' expressions of happiness lead to inferences about good team performance, whereas expressions of anger lead to inferences about poor team performance. Similarly, Liu et al. (2017) showed that a leader's positive affect significantly enhanced employees' psychological safety and thus promoted employees' voice behaviors.

Emotions serve as powerful social signals, allowing employees to infer their leaders' motivation and behavioral intentions (Dasborough & Ashkanasy, 2002; Keltner & Haidt, 1999). Importantly, emotions

signal whether a counterpart has prosocial motivations toward an individual. For example, Barasch et al. (2014) showed bystanders evaluated a donor more favorably when they believed the donation was motivated by genuine positive feelings from helping others. Related research has found that individuals who signal a diminished capability to empathize with others, through actions such as surface acting or emotional reactions inappropriate for the given context (e.g., laughing at a funeral), are perceived as less warm and benevolent than individuals viewed as having a high capability to experience and understand the emotions of others (Ashforth & Humphrey, 1993; Grandey, 2003; Gross, 1998; van Kleef, 2016). Inferences of a counterpart's emotional experience significantly shape perceptions of their prosocial motivation (Aknin et al., 2018; van Kleef & Lelieveld, 2022), and individuals perceived as incapable of experiencing emotions, such as empathy, are judged as less benevolent.

AI agents demonstrate high capability across many domains (e.g., beating humans in chess, go, and poker), but they are perceived as deficient in their capacity to experience emotions (Bigman & Gray, 2018; H. M. Gray et al., 2007; K. Gray & Wegner, 2012; Huebner, 2010; Kramer et al., 2018; Searle, 1980). AI can recognize and mimic human emotions using techniques such as convolutional neural networks for facial expression analysis (Kajale et al., 2023; Ng et al., 2023). However, AI's identification and expression of emotions is based on pattern recognition and data analysis rather than genuine emotional comprehension through "the experience of a sentient being" (Sajjad et al., 2023, p. 820). While AI can simulate an understanding of emotions, it lacks the capacity to experience emotions in the way that humans do, and humans recognize this limitation. Given that the capacity to experience emotions is seen as essential for humans to empathize with others and engage in benevolent behaviors, such as altruistic helping (Gaertig et al., 2019; Hawk et al., 2011; Keltner et al., 2006; R. B. Lount, 2010), we anticipate that the incapability of AI to empathize with others will harm the extent to which it is viewed as benevolent compared to a human manager.

Hypothesis 1: AI managers are perceived as less benevolent than human managers.

AI Management and Trust

Perceived benevolence plays a crucial role in fostering trust within organizational settings (Mayer & Davis, 1999; Mayer et al., 1995). Research has consistently demonstrated that perceived benevolence of leaders, characterized by the demonstration of goodwill and the genuine concern for the well-being of employees, significantly influences the emotional bond and trust between managers and subordinates (Chen et al., 2014; Dirks & Ferrin, 2002; Hiller et al., 2019; Legood et al., 2021; Tomlinson et al., 2020). Individuals also frequently infer high ability and integrity in management, which has made benevolence a key antecedent of trust building between human managers and subordinates (Fletcher, 2001; Kark et al., 2003; Lapidot et al., 2007; Miller, 2015; Zucker, 1986).

We anticipate the incapability of AI to authentically experience emotions may harm perceived benevolence and thus impede the development of trust between a human subordinate and an AI manager for two reasons. First, prior work suggests that subordinates are more trusting of leaders who are perceived to authentically

experience emotions (Campagna et al., 2016; Caza et al., 2015; Côté et al., 2013; Krumhuber et al., 2007). Since AI is unable to experience emotions and humans intuit this limitation (Bigman & Gray, 2018; K. Gray & Wegner, 2012; H. M. Gray et al., 2007; Huebner, 2010; Kramer et al., 2018; Sajjad et al., 2023; Searle, 1980), trust in AI management should be lower than trust in human management. Second, emotional experience is seen as integral for a genuine desire to help others (Ashforth & Humphrey, 1993; Barasch et al., 2014; Grandey, 2003; Gross, 1998; van Kleef, 2016), so the lack of an ability to experience emotions should reduce the perceived benevolence of AI management compared to human management. Given that benevolence is a primary driver of trust in leadership (Fletcher, 2001; Kark et al., 2003; Lapidot et al., 2007; Miller, 2015; Zucker, 1986), we anticipate the lower perceived benevolence of AI management, versus human management, will harm trust and that perceptions of benevolence mediate the relationship between AI, versus human, management and trust.

Hypothesis 2: AI managers are trusted less than human managers, and perceptions of benevolence mediate this relationship.

Empathy Demand

The relationship between AI, versus human, management, perceived benevolence, and trust is likely to be moderated by whether the situation is a high or low empathy demand context. In line with prior work, we define empathy as an affective response where a person vicariously experiences the emotional state of another individual (M. H. Davis, 2018; Hawk et al., 2011; Hoffman, 1985, 2008; Loewenstein & Small, 2007), and we conceptualize empathy demand as the extent to which an individual wants a counterpart to experience the emotions they are feeling. Given that empathy is closely associated with helping behavior (Batson & Shaw, 1991; Batson et al., 1981; Bohns & Flynn, 2021; Coke et al., 1978), empathy demand is likely to be high in contexts where an individual wants sympathy and help from their counterpart, such as when the individual experiences an adverse event, distress, or has little control.

We anticipate that in high empathy demand contexts, the desire for empathetic support will cause individuals to attend more to the emotional capabilities of their counterparts, especially in manager-to-subordinate relationships where managers have control over the individual and the ability to improve or worsen the individual's situation. During emotionally challenging times, employees seek affective connection and compassionate management (Kanov et al., 2004; Scott et al., 2010; Toegel et al., 2013). In such contexts, given that artificial intelligence is perceived as devoid of emotional experience (Bigman & Gray, 2018; K. Gray & Wegner, 2012; H. M. Gray et al., 2007), a human manager's greater capability to empathize and share employees' emotional experience will likely cause them to be perceived as higher in benevolence and more deserving of trust than an AI manager. However, in low empathy demand contexts, the relative emotional capabilities of AI and human management will be less salient to employees. That is, the greater capability of a human manager, versus an AI manager, to experience emotions will be more apparent in a high empathy demand context than a low empathy demand context, and we postulate that the extent to which a situation calls for empathy will

moderate the effects of AI management on perceived benevolence and trust. We summarize our theoretical model in Figure 1.

Hypothesis 3: The extent to which the context requires empathy will moderate the indirect relationship between AI management and trust via perceived benevolence, such that this indirect effect will be stronger when the context requires higher empathy versus lower empathy.

Overview of Current Work

Our work investigates the relationship between AI management, perceived benevolence, and trust, and the effect that empathy demand context has on this relationship. Across a field study and three experiments, we identify benevolence as a robust mediator between AI management and trust. In Study 1, we investigate and compare the mediating effects of ability, benevolence, and integrity in a preregistered field study by surveying delivery riders under AI management with and without the presence of a human manager. In Study 2, we conduct an experiment and manipulate whether the manager is a human or AI to establish causality and enhance internal validity. We conceptually replicate the delivery rider scenario in Study 1 and examine the important and unique role that benevolence plays in trust of AI management.

In Studies 3 and 4, we directly test the role that empathy demand context plays in trust of AI management. In Study 3, we randomly assign participants to an AI manager or a human manager, orthogonally manipulate whether or not the context requires high levels of empathy, and then measure if participants decide to keep their current type of manager or switch to the alternative type of manager. In Study 4, we test our full theoretical model linking AI management's lack of emotions to perceived benevolence and trust and the moderating role of empathy demand context.

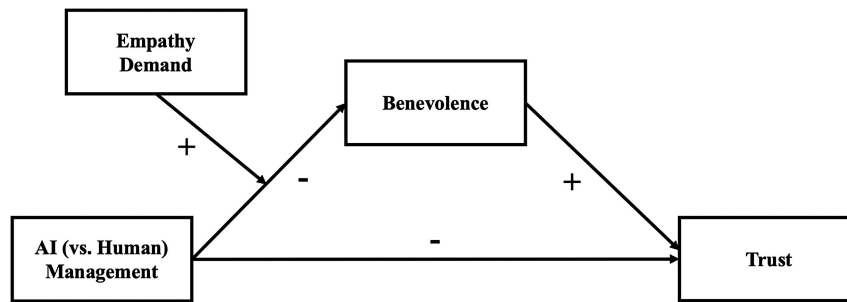
The procedures of all studies adhered to the ethical guidelines for the treatment of human participants and were approved by the Human Research Ethics Committee of the Hong Kong University of Science and Technology (HREP-2023-0056, "AI and Trust").

Transparency and Openness

We adhered to the *Journal of Applied Psychology* methodological checklist. We preregistered all studies and decided sample sizes in advance based on previous research (Dietvorst et al., 2015; Logg et al., 2019; Raveendhran & Fast, 2021). Data were analyzed using Jamovi 2.2.2 and Stata/SE 17.0. In each study, the sample size was neither increased nor decreased after preliminary data analysis. All measures, manipulations, and exclusions in every study are reported, and we include all preregistrations, data, syntax, and supplemental studies on Open Science Framework (OSF: <https://osf.io/49x7j/>).¹

¹ For each of our studies, consistent with our pre-registration we conducted ordinary least squares regression analysis and *t*-tests. For the *t*-tests, we report the beta value to aid in the comparison of our results within and across studies.

Figure 1
Theoretical Model



Note. AI = artificial intelligence.

Study 1: Delivery Riders' Trust of AI Management

Method

Sample Background

We examined delivery riders for the major online food delivery service platforms in mainland China. China's online food delivery industry is estimated to have a market size of U.S.\$153 billion as of the end of 2022, with approximately 521 million online users (China Internet Network Information Center, 2023; Liu, 2023). Meituan and Ele.me are the two dominant food delivery platforms in mainland China, and as of 2022, more than 7 million delivery riders earn their income through these two platforms (Ele.me, 2022; Meituan, 2023).

There are two main types of delivery riders who work for these services, with the major difference being the way they are managed (i.e., AI management or AI + human management). One type of riders are independent contract workers who are solely managed by the platform's AI system, similar to platforms such as Uber (Sun et al., 2023). For these riders, the AI system automatically provides them with delivery orders, manages their performance (e.g., fining riders for a late delivery or customer complaints), and even permanently blocks their accounts if they consistently ignore order assignments from the AI system or frequently delay deliveries. These riders do not have a human manager.

The other type of riders are managed by the platform's AI system while also being supported by a human manager. For these riders, the AI system automatically provides them with delivery orders and manages their performance, while the human station manager is responsible for holding daily morning meetings to provide coaching and training, overseeing the AI system's assignment of orders (e.g., when the AI system recommends an unreasonable order for a rider, the station manager can make changes), and helping employees appeal unreasonable fines.

Participants

We aimed to recruit 400 delivery riders from China, with 200 under AI management and 200 under AI + human management. A total of 400 delivery riders completed the study (369 male, $M_{\text{age}} = 29.92$ years, $SD = 8.29$).

Design and Procedure

From June 21 to June 29, 2023, we had research assistants distribute surveys to delivery riders taking breaks between orders in two Chinese cities, Linyi and Shenzhen. We gave riders a soda in exchange for their participation. Since delivery riders are on a tight schedule, we designed the survey to fit on the front and back of a single sheet of paper that can be completed in 3 min or less. We present the survey that was administered to the riders, in simplified Chinese, as well as the English translation in the online Supplemental Material.

In the survey, we first asked participants which type of riders they are: if they work for the platform as a crowdsourced rider (AI management) or also have a station manager as an outsourced rider (AI + human management). Next, we asked participants to rate the management on the following three items (7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*): "I will let platform management have any influence over issues that are important to me"; "I am satisfied with the level of control that platform management has over me"; and "I would trust platform management's decisions." We had participants rate the "platform management" to keep the scales consistent regardless of if individuals are evaluating the AI management or the AI + human management. We combined these items into an index of trust (adapted from Mayer & Davis, 1999; $\alpha = .87$).

We then asked participants to rate the management on the following items (adapted from Mayer & Davis, 1999; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*): "capable"; "competent"; "qualified"; "kind"; "warm"; "caring"; "fair"; "consistent"; "principled." We combined the first three items into an index of ability ($\alpha = .90$), the middle three items into an index of benevolence ($\alpha = .93$), and the final three items into an index of integrity ($\alpha = .90$). We randomly distributed six versions of surveys with the order of the three indices randomized to avoid order effects.

On the back page of the survey, we had participants complete our control variables. Before data collection, we spoke with 17 delivery riders and three station managers in mainland China to better understand the different types of riders. These were exploratory conversations to inform our subsequent data collection, and none of these individuals completed surveys. Based on the results of these interviews, we collected 19 control variables to account for differences between the two types of riders besides management type. In other words, we controlled for differences in workplace attributes which

might influence ratings of management but were not driven by the presence of a human manager. For example, we accounted for differences in rider characteristics (e.g., tenure, monthly income, gender, etc.), job characteristic (e.g., job autonomy, training, etc.), and potential variance due to survey collection (e.g., survey order, collection date, etc.). We provide explanations for the selection and measurement of control variables in the online [Supplemental Material](#).

We conducted confirmatory factor analyses using Jamovi 2.2.2 to examine the discriminant validity among our five main variables (i.e., rider type, ability, benevolence, integrity, and trust). We find that the five-factor model is a better fit with the original data, $\chi^2(56) = 83.6$, $\chi^2/df = 1.493$, comparative fit index, CFI = 0.993, Tucker–Lewis index, TLI = 0.990, standardized root-mean-square residual, SRMR = 0.022, root-mean-square error of approximation, RMSEA = 0.035, than the competing models (i.e., four-factor models, three-factor model, and one-factor model), indicating good discriminant validity among variables. We summarize our results in [Table 1](#).

The survey was administered in simplified Chinese to all delivery riders. All items were first developed in English and later translated into simplified Chinese following the back translation approach recommended by [Brislin \(1986\)](#). Specifically, the first author, who is bilingual in Chinese and English, translated the English items into Chinese and then a bilingual research assistant who was blind to the study's hypotheses back translated the Chinese items into English and the second author supervised and confirm consistency in meaning of items ([Klotz et al., 2023](#)).

Results

We conducted our regression analyses using Stata/SE 17.0. Following the recommendations of [Becker \(2005\)](#) and [Berneth and Aguinis \(2016\)](#), we report our results both without and with trimmed control variables. Four control variables (i.e., platform, alternative order, age, and survey order) did not change the significance level of any of the results and did not correlate with our dependent variables, so we report the results below without controlling for them to maximize statistical power and offer the most interpretable results. For transparency, we also report our results with full control variables in the online [Supplemental Material](#). We summarize our results in [Tables 2 and 3](#), and depict the key results in [Figure 2](#).

Table 1
Confirmatory Factor Analyses for Study 1

Model	χ^2	df	CFI	TLI	SRMR	RMSEA
Five-factor model	83.6	56	0.993	0.990	0.022	0.035
Four-factor model ^a	708	60	0.826	0.774	0.101	0.164
Four-factor model ^b	594	60	0.857	0.814	0.067	0.149
Four-factor model ^c	722	60	0.823	0.770	0.105	0.166
Three-factor model	1,252	63	0.682	0.606	0.111	0.217
One-factor model	1,575	65	0.596	0.515	0.120	0.241

Note. The theoretical model was a five-factor model (type, trust, ability, benevolence, integrity) and five competing models were constructed: ^aa four-factor model (type, trust, ability + benevolence, integrity), ^ba four-factor model (type, trust, ability + integrity, benevolence), ^ca four-factor model (type, trust, benevolence + integrity, ability), a three-factor model (type, trust, ability + benevolence + integrity), and a one-factor model (type + trust + ability + benevolence + integrity). CFI = comparative fit index; TLI = Tucker–Lewis index; SRMR = standardized root-mean-square residual; RMSEA = root-mean-square error of approximation.

Trust

We find that ratings of trust are significantly lower in the AI management group ($M = 3.62$, $SD = 1.78$) than in the AI + human management group ($M = 4.33$, $SD = 1.79$), $\beta = -0.71$, $SE = 0.18$, $p < .001$, $\eta^2 = 0.04$. Our result remains significant, $\beta = -0.47$, $SE = 0.23$, $p = .047$, $\eta^2 = 0.01$, when we include controls (see [Table 3](#) for the coefficients for the control variables).

Benevolence

Participants rate the platform management as significantly lower in benevolence in the AI management group ($M = 3.44$, $SD = 1.82$) than in the AI + human management group ($M = 4.90$, $SD = 1.75$), $\beta = -1.46$, $SE = 0.18$, $p < .001$, $\eta^2 = 0.14$. Our result remains significant, $\beta = -1.12$, $SE = 0.23$, $p < .001$, $\eta^2 = 0.08$, when we include controls.

Ability

Participants rate platform management as significantly lower in ability in the AI management condition ($M = 4.79$, $SD = 1.71$) than in the AI + human management condition ($M = 5.21$, $SD = 1.50$), $\beta = -0.42$, $SE = 0.16$, $p = .01$, $\eta^2 = 0.02$. Our result is no longer significant, $\beta = -0.07$, $SE = 0.21$, $p = .750$, $\eta^2 = 0.00$, when we include the control variables.

Integrity

Participants rate the platform management as significantly lower in integrity in the AI management condition ($M = 4.35$, $SD = 1.75$) than in the AI + human management group ($M = 4.77$, $SD = 1.72$), $\beta = -0.42$, $SE = 0.17$, $p = .017$, $\eta^2 = 0.01$. Our result is not significant, $\beta = 0.17$, $SE = 0.23$, $p = .446$, $\eta^2 = 0.00$, when we include the control variables.

Mediation

We conducted 10,000 simulation bootstrap analyses ([Hayes & Preacher, 2014](#); [Preacher & Hayes, 2004, 2008](#)) to investigate whether ability, benevolence, and integrity mediate the relationship between management type and participants' trust. More specifically, we examined trust as the dependent variable and management type (AI vs. AI + human) as the independent variable, with ability, benevolence, and integrity as simultaneous mediators. We find significant indirect effects of ability (IE = -0.05 , 95% CI [-0.13 , -0.00]), benevolence (IE = -0.59 , 95% CI [-0.83 , -0.38]), and integrity (IE = -0.07 , 95% CI [-0.17 , -0.01]) on the relationship between management type and trust. After including our control variables, we find a significant indirect effect of benevolence on the relationship between management type and trust (IE = -0.39 , 95% CI [-0.62 , -0.20]), but no significant indirect effects for ability (IE = -0.01 , 95% CI [-0.06 , 0.05]) and integrity (IE = 0.01 , 95% CI [-0.04 , 0.07]).

Discussion

In Study 1, we survey delivery riders, and we find that trust of management is significantly lower in the AI management group than in the AI + human management group. We also find that delivery

Table 2
Means, Standard Deviations, and Correlations for Study 1

Variable	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Rider type ^a	0.50	0.50	—															
2. Trust	3.98	1.82	-.195*** N = 400	(.87)														
3. Ability	5.00	1.62	-.128* N = 400	.422*** N = 400	(.90)													
4. Benevolence	4.17	1.93	-.380*** N = 400	.566*** N = 400	.512*** N = 400	(.93)												
5. Integrity	4.56	1.75	-.119* N = 400	.437*** N = 400	.542*** N = 400	.493*** N = 400	(.90)											
6. Work tenure	4.00	1.55	.456*** N = 397	-.177*** N = 397	-.143** N = 397	-.250*** N = 397	-.213*** N = 397	—										
7. Prior work experience	0.43	0.50	.255*** N = 390	-.118* N = 390	-.098 N = 390	-.087 N = 390	-.119* N = 390	.307*** N = 389	—									
8. Job autonomy	4.55	1.94	.184*** N = 400	.296*** N = 400	.410*** N = 400	.220*** N = 400	.270*** N = 400	.078 N = 397	.056 N = 390	(.73)								
9. Order	42.04	14.17	.024 N = 380	.062 N = 380	.110* N = 380	.048 N = 400	.149** N = 380	.074 N = 377	-.051 N = 372	-.012 N = 380	—							
10. Monthly income	6376.05	2094.51	.068 N = 369	-.012 N = 369	.125* N = 369	.011 N = 369	.000 N = 369	.241*** N = 368	.130* N = 361	.080 N = 369	.425*** N = 355	—						
11. Training	2.95	1.37	-.369*** N = 397	.215*** N = 397	.165*** N = 397	.330*** N = 397	.292*** N = 397	-.279*** N = 396	-.156** N = 388	.011 N = 397	.124* N = 377	.011 N = 367	—					
12. Fine appeal	2.69	1.48	.015 N = 397	-.036 N = 397	-.059 N = 397	-.068 N = 397	-.129* N = 397	.022 N = 397	.010 N = 388	.035 N = 397	.049 N = 377	.069 N = 367	.160** N = 396	—				
13. Insurance claim	1.54	1.01	-.081 N = 397	.126* N = 397	.010 N = 397	.086 N = 397	.021 N = 397	-.089 N = 396	-.048 N = 388	.017 N = 397	-.068 N = 377	-.003 N = 367	.341*** N = 396	.145** N = 396	—			
14. Group identity	4.96	1.76	-.060 N = 396	.298*** N = 396	.403*** N = 396	.310*** N = 396	.393*** N = 396	-.009 N = 394	-.014 N = 387	.307*** N = 396	.202*** N = 376	.061 N = 366	.150** N = 394	-.014 N = 394	—			
15. Job meaning	4.53	1.94	-.146** N = 395	.444*** N = 395	.487*** N = 395	.509*** N = 395	.443*** N = 395	-.110* N = 392	-.084 N = 385	.433*** N = 395	.066 N = 376	.049 N = 365	.221*** N = 392	-.098 N = 392	.036 N = 392	456*** N = 393	—	
16. Social desirability	5.08	2.02	.045 N = 397	.248*** N = 397	.273*** N = 397	.182*** N = 397	.158** N = 397	.027 N = 394	-.017 N = 387	.363*** N = 397	.045 N = 377	-.025 N = 366	.111* N = 394	.007 N = 394	.000 N = 377	.206*** N = 395	.280*** N = 394	—
17. Gender ^b	0.98	0.15	-.017 N = 378	.097 N = 378	.054 N = 378	.037 N = 378	.052 N = 378	.045 N = 378	.030 N = 370	-.061 N = 378	.006 N = 361	.090 N = 353	-.007 N = 377	-.010 N = 377	.013 N = 377	.126* N = 376	-.092 N = 375	-.036 N = 376
18. Education 1	0.19	0.39	.109* N = 385	.025 N = 385	-.070 N = 385	-.106* N = 385	-.071 N = 385	.083 N = 385	.029 N = 377	.072 N = 385	-.086 N = 367	-.069 N = 360	-.122* N = 383	-.024 N = 383	.058 N = 383	.059 N = 382	.042 N = 380	.116* N = 382
19. Education 2	0.21	0.40	.033 N = 385	.028 N = 385	.002 N = 385	.035 N = 385	.076 N = 385	.026 N = 384	.095 N = 377	.011 N = 385	.033 N = 377	.039 N = 360	-.042 N = 383	.013 N = 383	-.017 N = 383	.102* N = 382	.098 N = 380	-.062 N = 382
20. Education 3	0.37	0.48	-.066 N = 385	-.110* N = 385	-.025 N = 385	-.022 N = 385	-.071 N = 385	.031 N = 384	-.042 N = 377	-.060 N = 385	-.065 N = 367	-.011 N = 360	.002 N = 383	.018 N = 383	.002 N = 383	-.101* N = 382	-.107* N = 380	-.089 N = 382
21. Education 4	0.18	0.38	-.045 N = 385	.048 N = 385	.058 N = 385	.021 N = 385	.027 N = 385	-.151*** N = 384	-.001 N = 377	-.020 N = 385	.096 N = 367	.098 N = 360	.102* N = 383	-.016 N = 383	-.048 N = 382	-.072 N = 380	-.069 N = 380	.018 N = 382
22. Education 5	0.05	0.22	.007 N = 385	.058 N = 385	.068 N = 385	.112* N = 385	.074 N = 385	.045 N = 384	-.125* N = 377	.025 N = 385	.067 N = 367	-.067 N = 360	.059 N = 383	.000 N = 383	.027 N = 383	.057 N = 382	.073 N = 380	.042 N = 382
23. Education 6	0.01	0.09	-.091 N = 385	.011 N = 385	.030 N = 385	.059 N = 385	.060 N = 385	-.114* N = 384	-.017 N = 377	-.010 N = 385	.007 N = 367	-.074 N = 360	.131* N = 383	.019 N = 383	-.046 N = 382	.001 N = 380	.067 N = 380	.083 N = 382
24. Date 1	0.11	0.32	.063 N = 400	.191*** N = 400	-.029 N = 400	.027 N = 400	.014 N = 400	.030 N = 400	.008 N = 397	.078 N = 400	-.067 N = 380	-.207*** N = 369	.025 N = 397	.027 N = 397	-.024 N = 397	.004 N = 396	.024 N = 395	.025 N = 397

(table continues)

Table 2 (continued)

Variable	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
25. Date 2	0.33	0.47	.053 N = 400	-.080 N = 400	.075 N = 400	.027 N = 400	.211*** N = 400	-.143*** N = 397	-.128* N = 390	.000 N = 400	.213*** N = 380	-.196*** N = 369	.034 N = 397	-.113* N = 397	-.086 N = 397	.237*** N = 396	.070 N = 395	-.052 N = 397
26. Date 3	0.06	0.24	-.217*** N = 400	.124* N = 400	.055 N = 400	.171*** N = 400	.104* N = 400	-.074 N = 397	-.081 N = 390	-.088 N = 400	.149*** N = 380	-.097 N = 369	.139*** N = 397	-.002 N = 397	-.006 N = 397	.047 N = 396	.063 N = 395	.072 N = 397
27. Date 4	0.23	0.42	-.349*** N = 400	-.034 N = 400	-.019 N = 400	.044 N = 400	-.157*** N = 400	-.047 N = 397	.018 N = 390	-.080 N = 400	-.208*** N = 380	.140*** N = 369	.077 N = 397	.024 N = 397	.095 N = 397	-.159*** N = 396	-.046 N = 395	-.031 N = 397
28. Date 5	0.15	0.36	.221*** N = 400	-.020 N = 400	-.028 N = 400	-.041 N = 400	-.102* N = 400	.148*** N = 397	.165*** N = 390	.040 N = 400	-.070 N = 380	.219*** N = 369	-.143*** N = 397	.027 N = 397	.020 N = 397	-.002 N = 396	-.096 N = 395	.034 N = 397
29. Date 6	0.09	0.28	.257*** N = 400	-.094 N = 400	-.073 N = 400	-.216*** N = 400	-.081 N = 400	.137*** N = 397	.015 N = 390	.048 N = 400	.006 N = 380	.158*** N = 369	-.132*** N = 397	.065 N = 397	-.036 N = 397	-.130* N = 396	.003 N = 395	.018 N = 397
30. Date 7	0.02	0.15	.017 N = 400	-.035 N = 400	-.003 N = 400	-.037 N = 400	-.017 N = 400	.022 N = 397	.038 N = 390	.014 N = 400	-.060 N = 380	.013 N = 369	-.007 N = 397	.043 N = 397	.086 N = 397	-.131** N = 396	-.015 N = 395	-.040 N = 397
31. Location ^c	0.50	0.50	.000 N = 400	-.104* N = 400	-.073 N = 400	-.112* N = 400	-.256*** N = 400	.145*** N = 397	.140*** N = 390	-.008 N = 400	-.240*** N = 380	.374*** N = 369	-.107* N = 397	.098 N = 397	.101* N = 397	-.239*** N = 396	-.097 N = 395	.005 N = 397
Variable	M	SD	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
17. Gender ^b	0.98	0.15	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
18. Education 1	0.19	0.39	-.110* N = 369	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
19. Education 2	0.21	0.40	.081 N = 369	-.244*** N = 385	—	—	—	—	—	—	—	—	—	—	—	—	—	
20. Education 3	0.37	0.48	.012 N = 369	-.369*** N = 385	-.391*** N = 385	—	—	—	—	—	—	—	—	—	—	—	—	
21. Education 4	0.18	0.38	-.017 N = 369	-.224*** N = 385	-.237*** N = 385	-.359*** N = 385	—	—	—	—	—	—	—	—	—	—	—	
22. Education 5	0.05	0.22	.037 N = 369	-.109* N = 385	-.116* N = 385	-.175*** N = 385	-.106* N = 385	—	—	—	—	—	—	—	—	—	—	
23. Education 6	0.01	0.09	.014 N = 369	-.043 N = 385	-.045 N = 385	-.068 N = 385	-.041 N = 385	-.020 N = 385	—	—	—	—	—	—	—	—	—	
24. Date 1	0.11	0.32	-.050 N = 378	.008 N = 385	-.009 N = 385	-.018 N = 385	.058 N = 385	-.047 N = 385	-.033 N = 385	—	—	—	—	—	—	—	—	
25. Date 2	0.33	0.47	-.036 N = 378	-.002 N = 385	.021 N = 385	-.067 N = 385	.027 N = 385	.042 N = 385	.062 N = 385	-.250*** N = 400	—	—	—	—	—	—	—	
26. Date 3	0.06	0.24	-.028 N = 378	-.045 N = 385	.023 N = 385	-.028 N = 385	-.013 N = 385	.135*** N = 385	-.023 N = 385	-.093 N = 400	-.179*** N = 400	—	—	—	—	—	—	
27. Date 4	0.23	0.42	.040 N = 378	-.022 N = 385	-.092 N = 385	.112* N = 385	.005 N = 385	-.031 N = 385	-.046 N = 385	-.198*** N = 400	-.382*** N = 400	-.142*** N = 400	—	—	—	—	—	
28. Date 5	0.15	0.36	.066 N = 378	.051 N = 385	.030 N = 385	-.019 N = 385	-.033 N = 385	-.032 N = 385	-.038 N = 385	-.154*** N = 400	-.297*** N = 400	-.111* N = 400	-.236*** N = 400	—	—	—	—	
29. Date 6	0.09	0.28	-.010 N = 378	.011 N = 385	.041 N = 385	.000 N = 385	-.054 N = 385	-.030 N = 385	.075 N = 385	-.112* N = 400	-.215*** N = 400	-.080 N = 400	-.170*** N = 400	-.133** N = 400	—	—	—	
30. Date 7	0.02	0.15	.021 N = 378	-.023 N = 385	.016 N = 385	.039 N = 385	-.021 N = 385	-.033 N = 385	-.013 N = 385	-.055 N = 400	-.105* N = 400	-.039 N = 400	-.084 N = 400	-.065 N = 400	-.047 N = 400	—	—	
31. Location ^c	0.50	0.50	.081 N = 378	.016 N = 385	-.028 N = 385	.096 N = 385	-.059 N = 385	-.076 N = 385	-.027 N = 385	-.329*** N = 400	-.694*** N = 400	-.258*** N = 400	.550*** N = 400	.428*** N = 400	.310*** N = 400	.118* N = 400	—	

Note. Numbers in parentheses are Cronbach's α coefficients (or Spearman-Brown coefficient) for each scale. *M* and *SD* are raw values. AI = artificial intelligence.

^a Dummy-coded variable: 0 = AI + human riders, 1 = AI riders. ^b Dummy-coded variable: 0 = female, 1 = male. ^c Dummy-coded variable: 0 = Linyi, 1 = Shenzhen.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 3
Regression Table for Study 1

Variable	Trust					
	Model 1		Model 2		Model 3	
	β	SE	β	SE	β	SE
Constant	0.931	(0.778)	1.046	(0.776)	0.118	(0.721)
Rider type ^a			−0.467*	(0.234)	−0.080	(0.224)
Ability					0.100	(0.067)
Benevolence					0.351***	(0.058)
Integrity					0.082	(0.061)
Work tenure	−0.201**	(0.065)	−0.153*	(0.069)	−0.084	(0.064)
Prior work experience	−0.165	(0.187)	−0.109	(0.188)	−0.139	(0.173)
Job autonomy	0.148**	(0.052)	0.167**	(0.053)	0.108*	(0.049)
Order	0.008	(0.008)	0.008	(0.008)	0.011	(0.007)
Monthly income	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)
Training	0.042	(0.073)	0.005	(0.075)	−0.061	(0.070)
Fine appeal	−0.047	(0.062)	−0.045	(0.062)	0.003	(0.058)
Insurance claim	0.253*	(0.100)	0.256*	(0.099)	0.205*	(0.091)
Group identity	0.112	(0.059)	0.102	(0.059)	0.051	(0.056)
Job meaning	0.249***	(0.057)	0.242***	(0.057)	0.081	(0.056)
Social desirability	0.084	(0.046)	0.087	(0.046)	0.065	(0.042)
Gender ^b	1.460*	(0.577)	1.454*	(0.574)	1.204*	(0.527)
Education 1	−0.026	(0.283)	−0.027	(0.282)	−0.239	(0.259)
Education 2	−0.179	(0.259)	−0.200	(0.258)	−0.330	(0.237)
Education 3	0.246	(0.307)	0.262	(0.306)	0.012	(0.282)
Education 4	0.033	(0.430)	0.064	(0.428)	−0.318	(0.395)
Education 5	−0.287	(0.946)	−0.408	(0.943)	−0.589	(0.863)
Date 1	−1.488***	(0.294)	−1.455***	(0.293)	−1.397***	(0.269)
Date 2	−0.007	(0.431)	−0.154	(0.435)	−0.476	(0.401)
Date 3	0.417	(0.941)	0.407	(0.936)	0.569	(0.857)
Date 4	0.486	(0.953)	0.648	(0.951)	0.726	(0.870)
Date 5	0.042	(0.959)	0.280	(0.961)	0.634	(0.881)
Date 6	−0.148	(0.994)	−0.013	(0.991)	0.041	(0.909)
Location ^c	−1.295	(0.946)	−1.429	(0.944)	−1.420	(0.863)
R ²	0.3727		0.3811		0.4878	

Note. SE = standard error; AI = artificial intelligence.

^a Dummy-coded variable: 0 = AI + human riders, 1 = AI riders.

^b Dummy-coded variable: 0 = Linyi, 1 = Shenzhen.

* $p < .05$. ** $p < .01$. *** $p < .001$.

^c Dummy-coded variable: 0 = female, 1 = male.

riders evaluate management as less benevolent in the AI management group than in the AI + human management, and that perceived benevolence mediates the relationship between management type and trust. These findings reveal that benevolence may be an important driver for whether individuals decide to trust AI management.

There are two main limitations of this study. First, although we tried to account for potential confounds with our control variables, given the nature of survey responses, there are potential concerns of reverse causality and omitted variables which could influence our results. Second, to keep the phrasing of the survey consistent across all participants (i.e., AI management riders & AI + human management riders), we had the drivers rate the “management” instead of their “manager.” We address these concerns in our next study.

Study 2: Benevolence as Mediator

In Study 2, we extend our investigation in several ways. First, we conceptually replicate our field study in a preregistered experiment to test the causality of our model. Second, we explore and compare whether the management framing (i.e., “manager” vs. “management”) will influence employees’ perceptions and trust in AI management.

Method

Participants

We recruited 400 full-time employed participants from the United States via Prolific Academic to participate in a scenario study in exchange for \$0.60. A total of 400 people completed the study (249 male, 149 female, 2 nonbinary, $M_{\text{age}} = 40.86$ years, $SD = 11.28$).

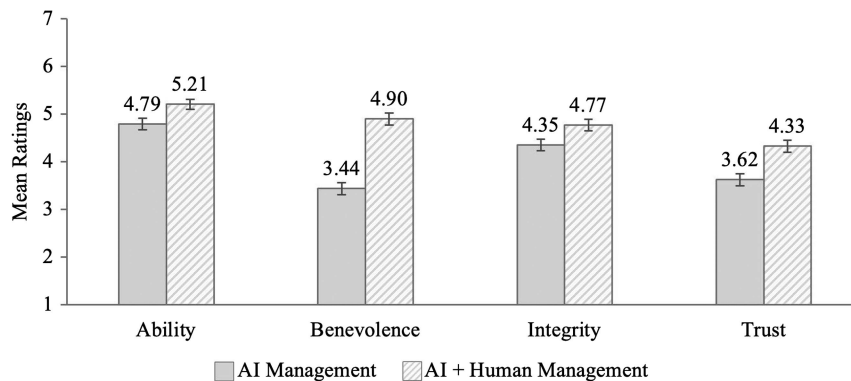
Design and Procedure

We used a 2 (AI vs. human manager) $\times$ 2 (manager vs. management framing) design. We randomly assigned participants to one of four between-subjects conditions: AI manager ($N = 103$) versus human manager ($N = 94$) versus AI management ($N = 101$) versus human management ($N = 102$).

Across all conditions, we asked participants to imagine that they are working as a delivery driver² for Company X, a food delivery

² In Study 1, we referred to delivery workers as ‘riders’ because it was the term that participants used to refer to themselves in China, since they primarily use motorbikes. In Study 2, we refer to delivery workers as ‘drivers’ because U.S. participants would be more familiar with that term.

Figure 2
Perceptions of Ability, Benevolence, Integrity, and Trust (Study 1)



Note. Standard error bars represent ± 1 SE. AI = artificial intelligence; SE = standard error.

platform, and their job is to pick up customer orders from restaurants and to deliver them, on time, to customers' homes.

In the AI conditions, we informed participants that their manager, Ari (Company X's AI-powered system), oversees their performance. In the human conditions, we informed participants that their manager, Ari (Company X's regional manager), oversees their performance. In both conditions, we then told participants that Ari gives them the orders that they need to complete, tracks if they complete the orders on time, pays them for the orders they complete, and deducts money from their paycheck if they receive a customer complaint.

Next, we manipulated whether participants rated the manager, Ari, or the management, Company X's management. In the manager conditions, we asked participants to rate Ari on the following three items (7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*): "If I had my way, I wouldn't let Ari have any influence over issues that are important to me" (reverse scored); "I would be willing to let Ari have complete control over my future in this company"; and "I would trust Ari's decisions." In the management conditions, we asked participants to complete the same three items but rate Company X's management (e.g., "I would be willing to let Company X's Management have complete control over my future in this company"). We combined these items into an index of trust (adapted from Mayer & Davis, 1999; $\alpha = .84$).

Next, we asked participants to rate Ari (in the manager conditions) or Company X's management (in the management conditions) on the following items (adapted from Mayer & Davis, 1999; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*): "Ari (Company X's management) is very capable of performing the job"; "Ari (Company X's management) has much knowledge"; "I feel very confident in the skills of Ari (Company X's management)"; "Ari (Company X's management) has specialized capabilities that can increase the company's performance"; "Ari (Company X's management) is well qualified"; "Ari (Company X's management) is very concerned about the welfare of others"; "The needs and desires of others are very important to Ari (Company X's management)"; "Ari (Company X's management) would not knowingly do anything to hurt others"; "Ari (Company X's management) really looks out for what is important to others"; "Ari (Company X's management) would try hard to help others";

"I would never have to wonder whether Ari (Company X's management) is honest"; "Ari (Company X's management) would be fair"; "The actions and judgment of Ari (Company X's management) are not very consistent" (reverse scored); "Sound principles guide the behavior of Ari (Company X's management)"; and "Ari (Company X's management) has a strong sense of justice." We combined the first five items into an index of ability ($\alpha = .95$), the middle five items into an index of benevolence ($\alpha = .95$), and the final five items into an index of integrity ($\alpha = .89$).

After collecting our dependent variables, we then had participants complete two manipulation checks. To test if our AI versus human manipulation was successful, we asked participants to complete the following two items: "Ari is Company X's regional manager" (reverse scored); and "Ari is Company X's AI powered system" ($p = .66$; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*). To measure if our manager versus management framing was effective, we asked participants to rate the following two items: "On the previous pages you rated Company X's management" (reverse scored) and "On the previous pages you rated Ari" ($p = .55$; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*).

At the end of the study, we asked participants to provide demographic information (i.e., age, gender) and any additional comments.

Results

We conduct our regressions analyzing the effects of management type (AI = 0.5, human = -0.5), management framing (manager = 0.5, management = -0.5), and the interaction between management type and framing using Stata/SE 17.0. We summarize our results in Tables 4 and 5, and depict the key results in Figure 3.

Manipulation Checks

The manipulation checks are successful. Participants in the AI conditions rated Ari as significantly more likely to be an AI-powered system than participants in the human conditions, $\beta = 2.59$, $SE = 0.16$, $p < .001$, $\eta^2 = 0.40$. We find no significant effect of manager versus management framing, $\beta = 0.27$, $SE = 0.16$, $p = .092$,

Table 4*Means, Standard Deviations, and Correlations for Study 2*

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9
1. AI ^a	0.01	0.50	—								
2. Framing ^b	−0.01	0.50	.025	—							
3. Trust	3.20	1.41	−.139**	.009	(.84)						
4. Ability	4.30	1.34	−.131**	.091	.687***	(.95)					
5. Benevolence	3.70	1.48	−.222***	.013	.730***	.755***	(.95)				
6. Integrity	3.92	1.24	−.068	.104*	.744***	.814***	.825***	(.89)			
7. Male	0.62	0.49	−.072	−.007	.043	.016	.051	.014	—		
8. Age	40.86	11.28	.030	−.010	−.044	−.053	−.021	−.023	−.050	—	
9. AI × Framing	0.01	0.25	−.016	.020	−.020	.027	−.051	−.005	−.068	.030	—

Note. *N* = 400. Numbers in parentheses are Cronbach's α coefficients for each scale. *M* and *SD* are raw values. AI = artificial intelligence.

^a AI = 0.5, human = −0.5. ^b Manager = 0.5, management = −0.5.

* $p < .05$. ** $p < .01$. *** $p < .001$.

$\eta^2 = 0.01$, and no significant interaction effect between management type and framing, $\beta = 0.59$, $SE = 0.32$, $p = .062$, $\eta^2 = 0.01$.

The second manipulation check confirms that participants were aware of the framing. Participants in the management conditions were significantly more likely to agree that they rated Company X's "management" on the previous page than participants in the manager conditions, $\beta = 2.50$, $SE = 0.13$, $p < .001$, $\eta^2 = 0.49$. We also find a significant effect of the AI versus human manipulation, $\beta = 0.36$, $SE = 0.13$, $p = .005$, $\eta^2 = 0.02$, where participants were more likely to agree that they rated Company X's "management" in the human conditions than in the AI conditions, and a significant interaction which reveals that this gap is larger in the manager conditions than in the management conditions, $\beta = 0.63$, $SE = 0.26$, $p = .014$, $\eta^2 = 0.02$.

Trust

We find trust is significantly lower in the AI conditions ($M_{AI \text{ manager}} = 3.00$, $SD = 1.43$, $M_{AI \text{ management}} = 3.02$, $SD = 1.38$) than in the human conditions ($M_{Human \text{ manager}} = 3.45$, $SD = 1.45$, $M_{Human \text{ management}} = 3.35$, $SD = 1.36$), $\beta = -0.39$, $SE = 0.14$, $p = .005$, $\eta^2 = 0.02$. We find no significant effect of framing, $\beta = 0.04$, $SE = 0.14$, $p = .791$, $\eta^2 = 0.00$, and no significant interaction between management type and framing, $\beta = -0.13$, $SE = 0.28$, $p = .651$, $\eta^2 = 0.00$.

Benevolence

We find that perceived benevolence is significantly lower in the AI conditions ($M_{AI \text{ manager}} = 3.33$, $SD = 1.55$, $M_{AI \text{ management}} = 3.43$, $SD = 1.42$) than in the human conditions ($M_{Human \text{ manager}} = 4.15$, $SD = 1.36$, $M_{Human \text{ management}} = 3.93$, $SD = 1.46$), $\beta = -0.66$, $SE = 0.15$, $p < .001$, $\eta^2 = 0.05$. We find no significant effect of framing, $\beta = 0.06$, $SE = 0.15$, $p = .689$, $\eta^2 = 0.00$, and no significant interaction between management type and framing, $\beta = -0.32$, $SE = 0.29$, $p = .266$, $\eta^2 = 0.00$.

Ability

We find that perceived ability is significantly lower in the AI conditions ($M_{AI \text{ manager}} = 4.28$, $SD = 1.37$, $M_{AI \text{ management}} = 3.97$, $SD = 1.44$) than in the human conditions ($M_{Human \text{ manager}} = 4.57$, $SD = 1.12$, $M_{Human \text{ management}} = 4.38$, $SD = 1.35$), $\beta = -0.36$, $SE = 0.13$, $p = .008$, $\eta^2 = 0.02$. We find no significant effect of framing, $\beta = 0.25$, $SE = 0.13$, $p = .059$, $\eta^2 = 0.01$, and no significant interaction between management type and framing, $\beta = 0.12$, $SE = 0.26$, $p = .643$, $\eta^2 = 0.00$.

Integrity

We find that perceived integrity is not significantly lower in the AI conditions ($M_{AI \text{ manager}} = 3.96$, $SD = 1.27$, $M_{AI \text{ management}} = 3.72$,

Table 5*Regression Table for Study 2*

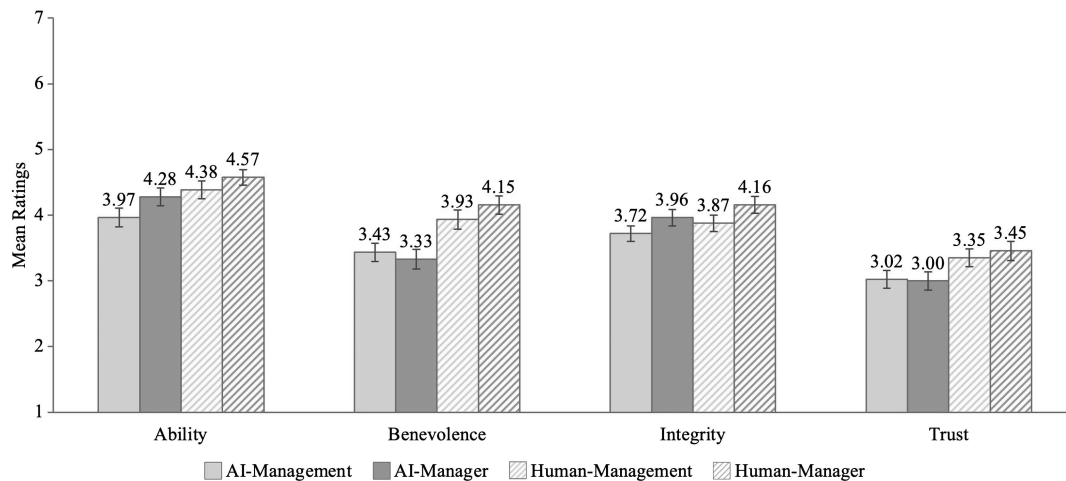
Variable	Ability		Benevolence		Integrity		Trust					
	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>
Constant	4.301***	(0.066)	3.713***	(0.073)	3.927***	(0.062)	3.207***	(0.070)	−0.220	(0.158)	−0.240	(0.159)
AI ^a	−0.356**	(0.133)	−0.661***	(0.145)	−0.175	(0.124)	−0.393**	(0.141)			−0.072	(0.094)
Framing ^b	0.252	(0.133)	0.058	(0.145)	0.262*	(0.124)	0.037	(0.141)			−0.133	(0.091)
AI × Framing	0.123	(0.266)	−0.323	(0.290)	−0.038	(0.247)	−0.127	(0.281)			−0.042	(0.180)
Ability									0.172**	(0.059)	0.174**	(0.060)
Benevolence									0.306***	(0.055)	0.281***	(0.058)
Integrity									0.395***	(0.074)	0.420***	(0.077)
<i>R</i> ²	0.0266		0.0525		0.0157		0.0199		0.6038		0.6065	

Note. *SE* = standard error; AI = artificial intelligence.

^a AI = 0.5, human = −0.5. ^b Manager = 0.5, management = −0.5.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Figure 3
Perceptions of Ability, Benevolence, and Integrity and Trust (Study 2)



Note. Standard error bars represent ± 1 SE. AI = artificial intelligence; SE = standard error.

$SD = 1.18$) than in the human conditions ($M_{\text{Human manager}} = 4.16$, $SD = 1.25$, $M_{\text{Human management}} = 3.87$, $SD = 1.24$), $\beta = -0.17$, $SE = 0.12$, $p = .158$, $\eta^2 = 0.01$. We find a significant effect of framing, $\beta = 0.26$, $SE = 0.12$, $p = .035$, $\eta^2 = 0.01$, with the manager framing being rated as higher in integrity than the management framing, but no significant interaction between management type and framing, $\beta = -0.04$, $SE = 0.25$, $p = .877$, $\eta^2 = 0.00$.

Mediation

We conducted a 10,000 simulation bootstrap analysis (Hayes & Preacher, 2014; Preacher & Hayes, 2004, 2008) to investigate whether ability, benevolence, and integrity mediate the relationship between management type and participants' trust. More specifically, we examined trust as the dependent variable and management type (AI vs. Human) as the independent variable, with ability, benevolence, and integrity as simultaneous mediators, controlling for management framing and the interaction between management type and framing. We find significant indirect effects of benevolence ($IE = -0.19$, 95% CI $[-0.32, -0.07]$) and ability ($IE = -0.06$, 95% CI $[-0.14, -0.01]$) on the relationship between management type and trust, but no significant indirect effect for integrity ($IE = -0.07$, 95% CI $[-0.19, 0.03]$).

Discussion

In Study 2, consistent with Study 1, we find that participants rate AI management as less benevolent and thus less trustworthy than human management. Specifically, we again find that perceptions of benevolence mediate the effect of AI management on trust.

Although the indirect effect is smaller than that for benevolence, we find that perceptions of ability also significantly mediate the relationship between AI management and trust. It is possible that participants' unfamiliarity with AI might cause them to underestimate its capabilities (Alan et al., 2014; Eslami et al., 2015; Linkov et al., 2017; Panella et al., 2003), and that participants' greater

familiarity with AI management in Study 1 lead to greater confidence in the ability of AI management than the participants in this study.

We examined the effect of having participants rate management or a specific manager to see if it could have influenced our results and found no significant effect of management framing on perceived trust, benevolence, and ability. Although we find a significant effect of message framing on perceived integrity, we find no significant interaction between management type and framing. Combined, the results indicate that message framing does not significantly influence our results.

Study 3: Empathy Demand as a Moderator

In Study 3, we test our theoretical model linking an AI manager's lack of emotional ability to perceived benevolence and trust and we investigate a moderator of trust in an AI manager; the extent to which individuals believe that the current context requires empathy. More specifically, we orthogonally manipulate if the situation calls for empathy or not and if the manager is human or AI to examine their effects on trust. We also conceptually replicate and extend the delivery rider context of Studies 1 and 2 and examine how the type of manager (AI vs. human) and empathy demands in the workplace affect if individuals decide to keep their current form of manager or switch to the other form.

Method

Participants

We recruited 600 full-time working participants from the United States via Prolific Academic to participate in a scenario study in exchange for \$0.50. A total of 600 people completed the study (386 male, 211 female, three nonbinary, $M_{\text{age}} = 41.66$ years, $SD = 11.30$).

Design and Procedure

We adopted a 2 (AI manager vs. human manager) $\times$ 2 (high empathy demand vs. low empathy demand) design and randomly assigned participants to one of four between-subjects conditions: human manager–high empathy demand ($N = 151$) versus human manager–low empathy demand ($N = 150$) versus AI manager–high empathy demand ($N = 148$) versus AI manager–low empathy demand ($N = 151$).

Across all conditions, we asked participants to imagine that they were working as a delivery driver for Company X, a food delivery platform, and their job was to pick up customer orders from restaurants and to deliver them, on time, to their homes. We then manipulated whether their manager, Ari, was a regional manager (i.e., human manager) or an AI-powered employee management system (i.e., AI manager). In all conditions, we told participants that Ari gave them the orders that they needed to complete, tracked if they completed the orders on time, paid them for the orders they completed, and deducted money from their paycheck if they received a customer complaint.

Next, we manipulated if the working context required high levels of empathy or not. In the high empathy demand conditions, we asked participants to imagine that they were completing their last order for the day. While driving to the customer's location, they were hit by someone who ran a red light. In the low empathy demand conditions, we informed participants that while driving to the customer's location, they did not encounter any issues.

We then manipulated whether participants' performance was normally evaluated by an AI manager or a human manager. Specifically, in the AI manager conditions, we told participants that their performance was normally evaluated by Company X's AI-powered employee management system, Ari. However, if they preferred, they could have their performance evaluated by a regional manager instead. In the human manager conditions, we told participants that their performance was normally evaluated by the regional manager, Ari. However, if they preferred, they could have their performance evaluated by Company X's AI-powered employee management system instead. In all conditions, we informed participants that both the regional manager and the AI-powered employee management system followed the same set of performance evaluation guidelines. We then asked participants if they wanted to keep their current manager or switch into the other group as a quasi-behavioral measurement of trust.

After collecting our dependent variable, we had participants complete two manipulation checks. To test if our AI manager versus human manager manipulation was successful, we asked participants to rate Ari on the following two items: "Ari is Company X's regional manager" (reverse scored); and "Ari is Company X's AI-powered employee management system" ($p = .93$; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*). To measure if our high empathy demand versus low empathy demand context was effective, we asked participants to rate the following four items: "Others should be empathetic in situations like this scenario"; "I would want others to treat me with compassion in this scenario"; "I would want others to be able to see things from my perspective in this type of scenario"; "I would want others to understand my emotions in this type of situation" ($\alpha = .95$; adapted from Joireman et al., 2006; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*).

At the end of the study, we asked participants to provide demographic information (age, gender, work experience) and any additional comments. Finally, we gave participants a completion link so that they could receive payment for the study.

Results

We conduct our regressions analyzing the effects of manager type (AI manager = 0.5, human manager = -0.5), context (high empathy demand = 0.5, low empathy demand = -0.5), and the interaction between manager type and context. We summarize our results in Table 6 and depict the key results in Figure 4.

Manipulation Checks

The manipulation checks are successful. Participants in the AI manager conditions rated Ari as significantly more likely to be an AI-powered system than participants in the human manager conditions, $\beta = 4.71$, $SE = 0.10$, $p < .001$, $\eta^2 = 0.78$. We find no significant effect of empathy demand context, $\beta = -0.08$, $SE = 0.10$, $p = .453$, $\eta^2 = 0.00$, and no significant interaction between manager type and context, $\beta = -0.07$, $SE = 0.21$, $p = .745$, $\eta^2 = 0.00$.

The second manipulation check confirms that participants in the high empathy demand conditions believed that the current context called for more empathy from the manager compared to participants in the low empathy demand conditions, $\beta = 1.23$, $SE = 0.10$, $p < .001$, $\eta^2 = 0.19$. We find no significant effect of the type of manager,

Table 6
Means, Standard Deviations, and Correlations for Study 3

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7
1. AI ^a	0.00	0.50	—						
2. Empathy demand ^b	0.00	0.50	-.007	—					
3. Trust	0.60	0.49	-.441***	-.149***	—				
4. Male	0.64	0.48	-.009	.032	-.028	—			
5. Age	41.66	11.30	-.011	.007	-.013	-.114**	—		
6. Work experience	20.41	11.57	-.027	-.002	.024	-.117**	.922***	—	
7. AI $\times$ Empathy Demand	0.00	0.25	-.003	-.003	-.179***	.016	.022	.009	—

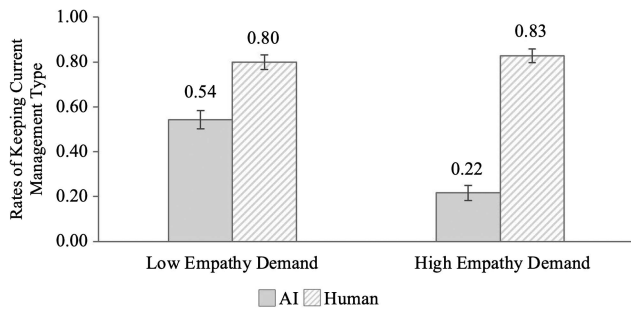
Note. $N = 600$. *M* and *SD* are raw values. AI = artificial intelligence.

^a AI manager = 0.5, human manager = -0.5. ^b High empathy demand = 0.5, low empathy demand = -0.5.

** $p < .01$. *** $p < .001$.

Figure 4

Moderating Effect of Empathy Demand on Trust of AI Versus Human Management (Study 3)



Note. Standard error bars represent ± 1 SE. AI = artificial intelligence; SE = standard error.

$\beta = -0.14$, $SE = 0.10$, $p = .177$, $\eta^2 = 0.00$, and no significant interaction between manager type and context, $\beta = 0.01$, $SE = 0.21$, $p = .954$, $\eta^2 = 0.00$.

Trust

We find significant effects of manager type, $\beta = -2.04$, $SE = 0.20$, $p < .001$, $OR = 0.13$, emotional context, $\beta = -0.64$, $SE = 0.20$, $p = .001$, $OR = 0.53$, and a significant interaction between manager type and emotional context, $\beta = -1.64$, $SE = 0.39$, $p < .001$, $OR = 0.19$, on participant decisions to keep their current form of manager. Participants were less likely to keep their current manager type in the AI manager conditions than in the human manager conditions, and this gap was larger in the high empathy demand conditions (82.8% kept the human manager vs. 21.6% kept the AI manager) than in the low empathy demand conditions (80.0% kept the human manager vs. 54.3% kept the AI manager).

Discussion

In Study 3, we replicated and extended our prior results in two ways. First, using a quasi-behavioral dependent measure, we find that when individuals are assigned to an AI manager or a human manager and given the option to switch to the alternative type of manager, individuals assigned to the AI manager are significantly more likely to want to switch. Second, we find a significant effect of emotional context on the extent to which individuals trust and decide to keep an AI manager. Employees distrust an AI manager more in high empathy demand working conditions compared to low empathy demand situations.

Although using a quasi-behavioral dependent measure is a strength of this study, a potential limitation is that it is a hypothetical scenario study. However, the consistency of results across our delivery worker field and scenario data in Studies 1 and 2 suggests that participants have no incentive to misrepresent their behavior in this scenario.

We used a delivery worker scenario in this study to gain a deeper understanding of the role of emotional context in our field study (Study 1) and its replication (Study 2). That is, by using a context similar to Studies 1 and 2, we are better able to understand the relationship between AI, versus human, manager, and trust we

observed in our prior studies. In Study 4, we test our full model in a different context to ensure the generalizability of our findings.

Study 4

In Study 4, we explore a common managerial scenario, an employee requesting time off, and test our full moderated mediation model. We postulate that in a context where individuals want empathy, AI's lack of emotions will be particularly salient and AI management will be perceived as less benevolent and trustworthy, compared to human management. In contrast, in a more emotionally neutral context, individuals will be less sensitive to AI's lack of emotions and more indifferent toward AI and human management.

Study 4: Pilot Study

We ran a preregistered pilot test of two scenarios to ensure that they were rated significantly different on the extent to which an employee would want their manager to treat them with empathy.

Method

Participants

We recruited 150 full-time working adults (89 male) online via Prolific to participate in a short study in exchange for \$0.30. The participants were, on average, 40.99 years old ($SD = 10.83$).

Design and Procedure

We randomly assigned participants into a high empathy demand condition ($N = 75$) or low empathy demand condition ($N = 75$).

In the high empathy demand condition, we asked participants to imagine that they have been working at Company X for 6 months. One day, during their afternoon break, they receive a text message informing them that their mother passed away. They are devastated and need to take next week off for the funeral and to help with the arrangements. In the low empathy demand condition, we asked participants to imagine that they have been working at Company X for 6 months and plan on taking next week off for vacation.

Then, using the same scale that we used in Study 3, we asked participants to rate to what extent they would want others to treat them with empathy in the context ($\alpha = .96$; adapted from Joireman et al., 2006).

Finally, we asked participants to provide their demographic information (i.e., age and gender).

Results and Discussion

We find that the extent to which participants would want others to treat them with empathy was significantly higher if they were planning to take time off because their mother had passed away ($M = 6.22$, $SD = 1.08$) than if they were planning to take time off for a vacation ($M = 4.49$, $SD = 1.65$), $\beta = 1.73$, $SE = 0.23$, $p < .001$, $\eta^2 = 0.28$.

Study 4: Main Study

Method

Participants

We recruited 800 participants with at least 1 year work experience from the United States via TurkPrime to participate in a scenario study in exchange for \$0.50. A total of 800 people completed the study (342 male, $M_{\text{age}} = 42.88$ years, $SD = 13.18$).

Design and Procedure

We used a 2 (AI management vs. human management) $\times$ 2 (high empathy demand vs. low empathy demand) design. We randomly assigned participants to one of four between-subjects conditions: human management–high empathy demand ($N = 202$) versus human management–low empathy demand ($N = 198$) versus AI Management–high empathy demand ($N = 203$) versus AI management–low empathy demand ($N = 197$).

Across all conditions, we asked participants to imagine that they have been working at Company X for 6 months. We informed participants that if they needed to take time off from Company X, they would be required to complete and submit a time-off form for management to evaluate and approve. The time-off form would require participants to provide their name, the date(s) of the absence, the time of absence, and the reason for their absence.

In the high empathy demand conditions, we informed participants that they were requesting time off next week because their mother had passed away and they would need to assist with the funeral arrangements. In the low empathy demand conditions, we informed participants that they were requesting time off next week for vacation.

Next, we manipulated whether the request was reviewed and approved by an AI manager or a human manager. Specifically, in the AI management conditions, we told participants that Company X's AI-powered employee management system reviewed their request and sent them an email confirming its approval, composed using historical internal emails. In the human management conditions, we told participants that a senior manager, Mike, had reviewed their request and sent them an email confirming it had been approved.

In all conditions, we showed the participant the email from their manager. To ensure external validity of the emails being potentially composed by AI, we generated these emails using ChatGPT. For the

email shown to participants in the high empathy demand conditions, we gave ChatGPT the following prompt: "Please write an email to your employees approving their leave of absence due to the death of their mother." For the email shown to participants in the low empathy demand conditions, we gave ChatGPT the following prompt: "Please write an email to your employees approving their leave of absence." We then revised the emails to keep the overall length and structure consistent across conditions. We present these stimuli in the online [Supplemental Material](#).

Next, across all conditions, we asked participants to rate Company X's management on the same measures of ability ($\alpha = .95$), benevolence ($\alpha = .95$), integrity ($\alpha = .89$), and trust ($\alpha = .75$; adapted from Mayer & Davis, 1999) we used in Study 2. The order of the ability, benevolence, and integrity measures was randomized across participants.

After collecting our dependent variables, we then had participants complete two manipulation checks. To test if our AI management versus human management manipulation was successful, we asked participants to rate Company X's Management on the following two items: "Your request was reviewed and approved by a Senior Manager, Mike" (reverse scored) and "Your request was reviewed and approved by Company X's AI-powered employee management system" ($p = .92$; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*). To measure if our high empathy demand versus low empathy demand context manipulation was effective, we asked participants to rate the following two items: "In this scenario, you want to take a leave because you want to take a vacation" (reverse scored) and "In this scenario, you want to take a leave because your mother passed away" ($p = .97$; 7-point Likert scale: 1 = *strongly disagree*, 7 = *strongly agree*).

At the end of the study, we asked participants to provide demographic information (age, gender, work experience) and any additional comments.

Results

We conduct our regressions analyzing the effects of management type (AI management = 0.5, human management = -0.5), context (high empathy demand = 0.5, low empathy demand = -0.5), and the interaction between management type and framing. We summarize our results in [Tables 7 and 8](#), and depict the key results in [Figure 5](#).

Table 7
Means, Standard Deviations, and Correlations for Study 4

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10
1. AI ^a	0.00	0.50	—									
2. Empathy demand ^b	0.01	0.50	.003	—								
3. Trust	4.48	1.25	-.055	.103**	(.75)							
4. Ability	5.52	1.08	-.077*	.132***	.654***	(.95)						
5. Benevolence	5.58	1.13	-.114**	.215***	.686***	.769***	(.95)					
6. Integrity	5.42	1.03	-.083*	.177***	.699***	.799***	.833***	(.89)				
7. Male	0.43	0.50	.000	-.026	-.107**	-.106**	-.128***	-.115**	—			
8. Age	42.88	13.18	.056	-.053	.027	.000	.081*	.049	-.036	—		
9. Work experience	21.28	12.27	.048	-.052	-.023	-.030	.053	.037	-.008	.898***	—	
10. AI $\times$ Empathy Demand	0.00	0.25	.013	.000	-.054	-.054	-.115**	-.067	-.012	-.016	-.018	—

Note. $N = 800$. Numbers in parentheses are Cronbach's α coefficients for each scale. *M* and *SD* are raw values. AI = artificial intelligence.

^a AI management = 0.5, human management = -0.5 . ^b High empathy demand = 0.5, low empathy demand = -0.5 .

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 8
Regression Table for Study 4

Variable	Ability		Benevolence		Integrity		Trust					
	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE
Constant	5.52***	(0.04)	5.58***	(0.04)	5.42***	(0.04)	4.48***	(0.04)	-0.54**	(0.17)	-0.60**	(0.17)
AI ^a	-0.16*	(0.08)	-0.25***	(0.08)	-0.17*	(0.07)	-0.14	(0.09)			0.05	(0.06)
Empathy demand ^b	0.28***	(0.08)	0.49***	(0.08)	0.36***	(0.07)	0.26**	(0.09)			-0.10	(0.06)
AI $\times$ Empathy Demand	-0.23	(0.15)	-0.51***	(0.15)	-0.27	(0.14)	-0.27	(0.18)			0.05	(0.12)
Ability									0.21***	(0.05)	0.21***	(0.05)
Benevolence									0.31***	(0.05)	0.32***	(0.05)
Integrity									0.39***	(0.06)	0.39***	(0.06)
R ²	0.0263		0.0723		0.0426		0.0166		0.5349		0.5368	

Note. AI = artificial intelligence; SE = standard error.

^a AI management = 0.5, human management = -0.5. ^b High empathy demand = 0.5, low empathy demand = -0.5.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Manipulation Checks

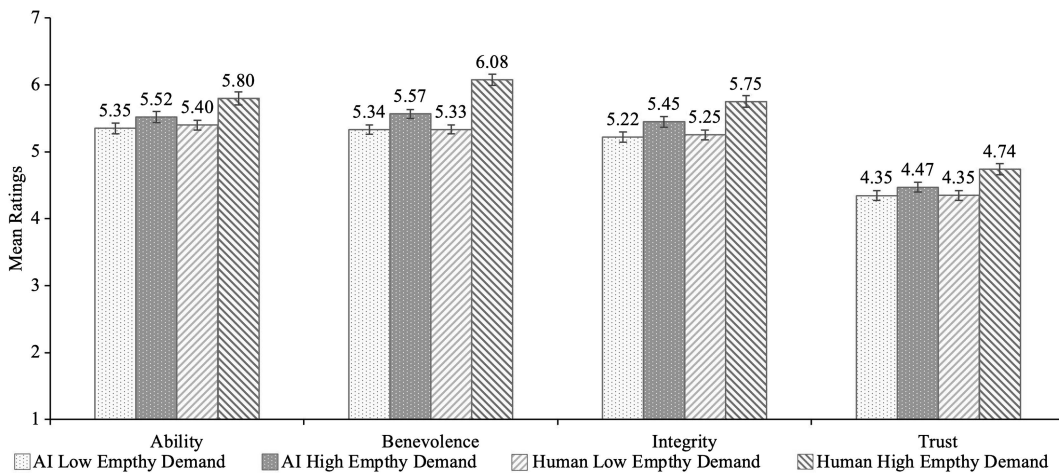
The manipulation checks are successful. Participants in the AI management conditions rated Company X's management as significantly more likely to be an AI-powered system than participants in the human management conditions, $\beta = 4.46$, $SE = 0.09$, $p < .001$, $\eta^2 = 0.76$. We find no significant effect of emotional context, $\beta = -0.04$, $SE = 0.09$, $p = .690$, $\eta^2 = 0.00$, and no interaction between management type and context, $\beta = 0.05$, $SE = 0.18$, $p = .781$, $\eta^2 = 0.00$.

The second manipulation check confirms that participants in the low empathy demand conditions were significantly more likely to perceive their leave as a vacation compared to participants in the high empathy demand conditions, $\beta = 5.11$, $SE = 0.06$, $p < .001$, $\eta^2 = 0.90$. We find no significant effect of management type, $\beta = -0.01$, $SE = 0.06$, $p = .896$, $\eta^2 = 0.00$, and no interaction between management type and context, $\beta = 0.01$, $SE = 0.12$, $p = .948$, $\eta^2 = 0.00$.

Trust

When we examine the effect of our manipulations on trust, we find no significant effect of management type, $\beta = -0.14$, $SE = 0.09$, $p = .123$, $\eta^2 = 0.00$, a significant effect of emotional context, $\beta = 0.26$, $SE = 0.09$, $p = .003$, $\eta^2 = 0.01$, and no significant interaction between management type and context, $\beta = -0.27$, $SE = 0.18$, $p = .127$, $\eta^2 = 0.00$. Trust is significantly higher in the human management–high empathy demand condition ($M = 4.74$, $SD = 1.18$) than in the AI management–high empathy demand condition ($M = 4.47$, $SD = 1.20$), $\beta = 0.27$, $SE = .12$, $p = .029$, $\eta^2 = 0.01$, but we find no significant difference in trust ratings in the human management–low empathy demand condition ($M = 4.35$, $SD = 1.38$) and the AI management–low empathy demand condition ($M = 4.35$, $SD = 1.19$), $\beta = 0.00$, $SE = 0.12$, $p = .990$, $\eta^2 = 0.00$.

Figure 5
Moderating Effect of Empathy Demand on Perceptions of ABI and Trust of AI Versus Human Management (Study 4)



Note. Standard error bars represent ± 1 SE. AI = artificial intelligence; SE = standard error; ABI = ability, benevolence, and integrity.

Benevolence

When we examine the effect of our manipulations on perceived benevolence, we find a significant effect of management type, $\beta = -0.25$, $SE = 0.08$, $p = .001$, $\eta^2 = 0.01$, a significant effect of emotional context, $\beta = 0.49$, $SE = 0.08$, $p < .001$, $\eta^2 = 0.05$, and a significant interaction between management type and context, $\beta = -0.51$, $SE = 0.15$, $p = .001$, $\eta^2 = 0.01$. We find that participants rate the management as significantly higher in benevolence in the human management–high empathy demand condition ($M = 6.08$, $SD = 0.93$) than in the AI management–high empathy demand condition ($M = 5.57$, $SD = 1.05$), $\beta = 0.51$, $SE = 0.11$, $p < .001$, $\eta^2 = 0.03$, but no significant difference in benevolence ratings in the human management–low empathy demand condition ($M = 5.33$, $SD = 1.22$) and the AI management–low empathy demand condition ($M = 5.34$, $SD = 1.14$), $\beta = 0.00$, $SE = 0.11$, $p = .995$, $\eta^2 = 0.00$.

Ability

When we examine the effect of our manipulations on perceived ability, we find a significant effect of management type, $\beta = -0.16$, $SE = 0.08$, $p = .030$, $\eta^2 = 0.01$, a significant effect of emotional context, $\beta = 0.28$, $SE = 0.08$, $p < .001$, $\eta^2 = 0.02$, and no significant interaction between management type and context, $\beta = -0.23$, $SE = 0.15$, $p = .127$, $\eta^2 = 0.00$. We find that participants rate the management as significantly higher in ability in the human management–high empathy demand condition ($M = 5.80$, $SD = 0.98$) than in the AI management–high empathy demand condition ($M = 5.52$, $SD = 1.08$), $\beta = 0.28$, $SE = 0.11$, $p = .009$, $\eta^2 = 0.01$, but no significant difference in ability ratings in the human management–low empathy demand condition ($M = 5.40$, $SD = 1.13$) and the AI management–low empathy demand condition ($M = 5.35$, $SD = 1.06$), $\beta = 0.05$, $SE = 0.11$, $p = .649$, $\eta^2 = 0.00$.

Integrity

When we examine the effect of our manipulations on perceived integrity, we find a significant effect of management type, $\beta = -0.17$, $SE = 0.07$, $p = .018$, $\eta^2 = 0.01$, a significant effect of emotional context, $\beta = 0.36$, $SE = 0.07$, $p < .001$, $\eta^2 = 0.03$, and no significant interaction between management type and context, $\beta = -0.27$, $SE = 0.14$, $p = .056$, $\eta^2 = 0.00$. We find that participants rate the management as significantly higher in integrity in the human management–high empathy demand condition ($M = 5.75$, $SD = 0.96$) than in the AI management–high empathy demand condition ($M = 5.45$, $SD = 1.01$), $\beta = 0.31$, $SE = 0.10$, $p = .002$, $\eta^2 = 0.01$, but no significant difference in integrity ratings in the human management–low empathy demand condition ($M = 5.25$, $SD = 1.04$) and the AI management–low empathy demand condition ($M = 5.22$, $SD = 1.01$), $\beta = 0.03$, $SE = 0.10$, $p = .743$, $\eta^2 = 0.00$.

Moderated Mediation

We examine whether the extent to which a given situation requires empathy will moderate the mediating effect of perceived ability, benevolence and integrity between manager type and trust. We conducted a 10,000 simulation bootstrap analysis (Hayes & Preacher, 2014; Preacher & Hayes, 2004, 2008), with trust as our dependent variable, ability, benevolence and integrity as our simultaneous mediator variables, management type (AI management = 0.5,

human management = -0.5) as our independent variable and empathy demand context (high empathy demand = 0.5, low empathy demand = -0.5) as the moderator. We find a significant index of moderated mediation for benevolence (index = -0.16 , 95% CI [-0.29 , -0.06]), but no significant index of moderated mediation for ability (index = -0.05 , 95% CI [-0.12 , 0.01]) or integrity (index = -0.11 , 95% CI [-0.23 , 0.003]). In the high empathy demand conditions, there is a significant indirect effect of benevolence (IE = -0.16 , 95% CI [-0.25 , -0.09]) on the relationship between management type and trust, but in the low empathy demand conditions the indirect effect is not significant (IE = 0.00 , 95% CI [-0.08 , 0.08]). We also find significant indirect effects for ability (IE = -0.06 , 95% CI [-0.11 , -0.01]) and integrity (IE = -0.12 , 95% CI [-0.21 , -0.04]) in the high empathy demand conditions, but no significant indirect effects for ability (IE = -0.01 , 95% CI [-0.06 , 0.04]) or integrity (IE = -0.01 , 95% CI [-0.09 , 0.07]) in the low empathy demand conditions.

Discussion

In Study 4, we examine how the extent to which individuals want empathy in a given context moderates the relationship between AI management and trust and test our full theoretical model. Our moderated mediation analysis reveals that emotional context and perceived benevolence play an important role in the relationship between AI management and trust. More specifically, the relationship between AI management, benevolence, and trust is stronger in a context which requires high levels of empathy than a situation which does not require high levels of empathy. Combined, these results support that emotional capability and perceived benevolence have significant effects on trust of AI management versus human management.

General Discussion

As society progresses into the fourth industrial revolution, AI-driven digital transformation in global companies is altering the dynamics of social interactions in the workplace, particularly between leaders and subordinates (Murray et al., 2021). Given that this is a recent shift, the existing AI and trust literature has focused mainly on the perspective of individuals who are the users of technology or how various characteristics of AI influence its adoption relative to other versions of AI (Bedué & Fritzsche, 2022; Choung et al., 2023; Langer et al., 2023; W. Wang & Benbasat, 2007).

Across four preregistered studies, we find that lower perceived benevolence harms trust of AI management. In a field study, we survey 400 delivery riders in mainland China and find that those who work only under AI perceive management as less benevolent and trust it less than those who also have a human manager. In our experiments, we test our causal model and find that perceptions of benevolence are lower under AI management than human management, which impairs trust in AI management, but that emotional context is an important moderator of these effects.³ In a high empathy demand context, where individuals want management to be able to experience their emotions, we find that AI managers

³ On OSF, we present six additional experiments where we replicate the effect of trust being lower with AI management than human management and that perceived benevolence mediates this effect.

are perceived as less benevolent than human managers, which has a negative effect on trust, but this effect is attenuated in a low empathy demand context. Together, our results underscore that emotional context and perceived benevolence (or lack thereof) play an important role in trust of AI management.

Theoretical Implications

Our studies make several important theoretical contributions. First, we extend interpersonal trust theory by deepening the understanding of AI management from a sociotechnical perspective. Research on technological adoption has traditionally conceptualized humans as voluntary users (Langer et al., 2023; W. Wang & Benbasat, 2007) and focused on factors related to the ability of technology to complement human operators and execute subordinate tasks (e.g., the perceived usefulness and ease-of-use; F. D. Davis, 1989; Lazar & Nelson, 2023). As AI assumes organizational positions previously held by humans, human-to-human models may enable us to better understand human-to-AI interactions. By extending the use of the interpersonal trust framework (ABI Model of Trust; Mayer et al., 1995) to understand the perspective of employees who are subordinate to AI systems, we deepen our understanding of the psychological factors which influence trust and adoption of AI management.

Second, we employ the interpersonal trust framework and the emotions as social information (EASI) model (Van Kleef, 2008; van Kleef & Côté, 2022) to bridge the trust and AI literatures and describe the important role that perceived benevolence plays in trust of AI management. A broad literature has demonstrated that a key antecedent of trust between human leaders and human subordinates is perceived benevolence (e.g., Chen et al., 2014; Dirks & Ferrin, 2002; Hiller et al., 2019), which is shaped by inferences of a leader's emotional capacity and capability to experience empathy (Ashforth & Humphrey, 1993; Grandey, 2003; van Kleef, 2016). Our findings suggest that AI management's inability to experience emotions, particularly empathy, has a detrimental effect on its perceived benevolence, and that benevolence is a key factor influencing trust of AI management.

Third, we make an important connection between the literatures on emotions and on algorithm and AI adoption in the workplace by highlighting the important moderating role that emotional context plays in the perceived benevolence and trust of AI management. We extend prior research, which has historically evaluated acceptance of AI based on the behavior of the system (e.g., if it makes an error, provides an explanation for its recommendations; Dietvorst et al., 2015; W. Wang & Benbasat, 2007), by considering the important roles that inferred empathy and emotional context play in employee trust of an emerging technology. Our findings reveal that empathy and emotional context, which have been shown to influence trust in human-to-human interactions (e.g., incidental emotions such as anger; Dunn & Schweitzer, 2005; van Kleef & Lelieveld, 2022), are important antecedents of trust in human-to-AI interactions in organizations.

Practical Implications

The lack of perceived benevolence of AI management may be an important impediment to its adoption. This provides a practical consideration for organizations, which should anticipate trust and

adoption issues when replacing roles and responsibilities historically occupied by humans with AI systems. Companies deploying AI management systems might consider simultaneously implementing supportive policies (e.g., more flexible leave policies, employee education and training programs) which could help elevate the perceived benevolence of management and potentially buffer against lost trust.

Our emotional context findings also reveal that employees will be less likely to trust and adopt AI management in situations which require a high level of empathy. Organizations should consider the nature of the work and the frequency of high empathy demand situations when deciding how much of their human management to replace with AI management. Alternatively, companies might want to selectively deploy AI management for low empathy demand situations while ensuring that human managers are still involved in high empathy demand situations. Organizations might also consider interventions which could increase the perceived capability of AI management to experience emotions, and we anticipate future research will clarify the efficacy of such interventions.

Limitations and Future Directions

Our findings emphasize the importance of considering multiple dimensions of trust when examining how employees perceive AI management. In our studies, benevolence is the key mechanism by which AI, versus human, management influences trust when humans are subordinate to AI. However, ability and integrity are also important antecedents of trust (Mayer & Davis, 1999; Mayer et al., 1995), and our results from Studies 1, 2, and 4 suggest that ability and integrity are likely to matter as well. Notably, human managers are generally perceived as higher in ability and integrity, compared to AI managers, but the effects of ability and integrity on trust of AI management are inconsistent across our studies. For example, we find a significant indirect effect of ability in Study 2, and significant indirect effects for both ability and integrity in Study 1 (when we do not include controls) and in the high empathy demand conditions in Study 4. While we cannot make any definitive claims since these samples are not directly comparable and vary on multiple dimensions, we anticipate that various contextual- and individual-level factors will influence the effects that perceived ability, benevolence, and integrity have on trust of AI and AI management.

In our studies, we only examined the perspective of individuals who are subordinate to AI in settings where AI has very strong capabilities (e.g., dispatching millions of delivery drivers). We anticipate that the perceived ability of an AI management system will have a stronger effect on trust for individuals who are senior to the AI system, who are essentially users (Dietvorst et al., 2015; Logg et al., 2019), and for employees who feel that they have greater expertise than management (Logg, 2019; Logg et al., 2019). While we find that employees generally prefer human management in a high empathy demand situation, it is also possible that employees with a greater need for autonomy might find the emotional indifference of AI management more appropriate and consistent with their own values (Raveendhran & Fast, 2021), which may lead them to evaluate AI management as higher in integrity and more trustworthy. Understanding how individual differences influence the perceived ability, benevolence, and integrity of AI management

will likely be important for ensuring the trust and adoption of such systems.

We did not directly measure the capability of AI management to experience emotions in our studies. Although prior work has shown that AI is unable to experience emotions and that individuals recognize this limitation (Bigman & Gray, 2018; Kajale et al., 2023; Ng et al., 2023; Sajjad et al., 2023), researchers have begun to explore factors, such as anthropomorphism, that might increase the perceived emotional experience of AI. Although some studies suggest that making AI managers more humanlike might increase trust, others suggest that attempting to make them too humanlike can lead to backlash (Mori et al., 2012; Schroeder & Epley, 2016; Schroeder & Schroeder, 2018). For example, K. Gray and Wegner (2012) found that human-looking robots are more likely to be viewed as possessing human minds, but this causes users to feel unsettled and assess the robots as creepy. In contrast, Antos et al. (2011) found that AI negotiators are perceived as more caring and likable when their emotional expressions are consistent with their negotiation strategies and behaviors. As AI technology continues to advance, it is possible that its emotional responses will be viewed as more accurate and authentic, and its perceived capability to experience emotions may increase as well.

While future research is needed to better understand how advancements such as anthropomorphism will influence the perceived emotional experience of AI, research suggests that a potential avenue for addressing the emotional limitations of AI is enhancing the emotional experience of the user. For example, Zhang et al. (2010) compared service robots with or without a humanlike facial configuration and found that anthropomorphic features of the robots led to positive emotional experiences for their human counterparts. Moreover, related research suggests that improving the emotional experience of the user, such as by making the avatar more attractive, can lead to downstream consequences such as making the AI agent appear to have greater social interaction capabilities and more positive personality traits (R. Khan & De Angeli, 2009; R. F. Khan & Sutcliffe, 2014). While it is possible that an AI agent that is physically attractive will still be seen as less benevolent and trustworthy than a human who is equally physically attractive, future research may reveal that improving the emotional experience of employees may be an effective method of mitigating the obstacles caused by AI management's lack of emotional capability.

Future research may also explore individual differences which might attenuate the perception that AI is deficient in its ability to experience emotions (Bigman & Gray, 2018; Kajale et al., 2023; Ng et al., 2023). Characteristics of employees (e.g., gender, age, race, culture) might influence the inferred emotional experience of AI management, its perceived benevolence, and the subsequent effect on trust. For example, members of highly industrialized cultures, who are more familiar with technology and the underlying processes by which it works, are less likely to anthropomorphize mechanical agents than individuals from less industrialized cultures (Epley et al., 2007). Future work might explore how cultural factors such as industrialization influence the perceived emotional capability and benevolence of AI. We anticipate that characteristics of the employee, the AI manager, and the context are all likely to play important roles in the trust of AI management.

Conclusion

AI management has vast potential. Yet, reaping these benefits is predicated on AI management being accepted and adopted. Employees view AI management as less benevolent than human management, which harms trust, particularly in situations where individuals want management to be capable of experiencing empathy. The extent to which AI management is perceived to be unable to genuinely feel empathy may forever be a major hurdle to individuals trusting and accepting AI-powered systems, and organizations should be mindful of emotional context in deciding which responsibilities they will grant to AI instead of human managers.

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